

Project Problem Proposal and Approval Form

Complete for Laboratory 2 and the ungraded Moodle approval checkpoint

Field	Team response
Team number	12
Working project title	Campus Lost and Found Management System
Members and student IDs	Shinny Kodie:202307950, Shirley Thiite:202300281, Becky Otsetswe:202302203, Thoriso Mochanang:202304714, Olga Gaborekwe: 202306791
Repository URL	https://github.com/Shirley-Thiite/csi473-project
Proposal tag/commit	v1.1
Date submitted	17/08/2026

Originality declaration: This proposal is based on our own investigated problem. It is not the University Service Hub lecturer case.

1. Problem statement (150-250 words)

Currently campus lost and found processes at universities are often inefficient and disorganized, relying on manual, physical drop-off points, social media posts, or word of mouth. This unstructured approach makes it difficult for students, faculty and staff to report lost items or find found belongings, leading to confusion, delays, and low recovery rates. There is no single reliable source of information, which can use many lost items to remain unclaimed, resulting in frustration and waste. This project will design a centralized, web-based platform to streamline reporting, searching, and claiming lost and found items within the campus community.

2. Evidence of the problem

Evidence/source	Date/origin	What it shows
Observations of physical lost and found drop-off locations on campus	7 August 2026 (Library), 7 August 2026 (Student Center)	Physical lost and found boxes in buildings are often overflowing and disorganized, making it difficult to search for items efficiently
Evidence from campus social media groups (WhatsApp and Facebook)	Ongoing campus observations from 7 August 2026 to 31 August 2026	Students frequently post lost items on WhatsApp and Facebook groups, highlighting the lack of a centralized system for lost and found on campus.

3. Goal and objectives

Item	Team response
Goal	The goal of this project is to design a software system that significantly increases the efficiency, transparency, and success rate of returning lost items to their owners by providing a structured, digital, and user-friendly platform.
Objective 1	Increase Recovery Rate: Improve the percentage of reported lost items successfully returned to their owners from the current estimated 28% to at least 60% within 6 months of deployment by providing a centralized database with search and matching capabilities, reducing reliance on inefficient manual methods.
Objective 2	Improve User Experience: Achieve a user satisfaction score of at least 4 out of 5 on post-interaction surveys by providing a clear, intuitive, and secure process for reporting, searching, and claiming items, thereby reducing frustration associated with the current system.
Objective 3	Reduce Time to Recover: Decrease the average time to reunite an owner with a lost item from the current estimated 5-7 days to under 48 hours by enabling real-time reporting, instant search, and automated notifications for potential matches.

4. Stakeholders

Stakeholder/role	Need	Concern	Evidence/access
Student, Staff, User	Report a lost item, search for a found item, submit a claim for an item, and receive updates.	Ease of use, security of personal data, speed of response, assurance that the claim process is fair.	Common campus experience—students frequently lose items (phones, laptops, ID cards, keys) and rely on fragmented methods (social media posts, bulletin boards, and physical lost-and-found offices).
Campus Administrator	Manage the system, verify and approve claims, oversee the handover process, generate reports.	System integrity, prevent fraudulent claims, ensuring a clear audit trail for all transactions.	Without formal verification, anyone could falsely claim expensive items.
Campus Security	Physically handle and store found items, coordinate with administrators, manage unclaimed items.	A clear, standardized process for intake, storage, and disposition to avoid confusion.	Current systems often lack clear procedures, leading to items being misplaced or disposed of improperly.
Campus IT/Developers	Ensure the system is reliable, secure, and maintainable.	System uptime, data security and privacy, scalability to handle campus-wide usage, cost-effectiveness.	General IT concerns any institutional system handling personal data.

5. Scope, assumptions and constraints

In scope	Out of scope
Reporting of lost and found items by authenticated users.	Integration with campus CCTV or security cameras.
A searchable database of lost and found items.	Integration with the university's central student database for automatic verification.
A claim process for users to request ownership of found items.	A mobile-native application, the initial scope is a responsive web-based system.
An administrative interface to manage items and review/approve/reject claims.	Multi-campus or inter-university support.

Assumptions	Constraints
A stable campus network is available for system access	Team capacity: Development is bounded by a small student team and the number of teaching weeks available, which limits the number of features that can be built to production quality.
Users will have access to a modern web browser	Infrastructure: The system must run on infrastructure the team can provide.
A designated campus office will act as the administrative owner of the system and manage the physical item handover	Time: the system must be designed and delivered as a semester-sized vertical slice within the CSI473 teaching timetable, not a full production rollout.

6. Proposed core workflow / vertical slice

Primary Actor: A student who has lost their laptop.

Task: Report the lost item and recover it when found.

1. **Report Item:** Student logs in and submits a report as either lost or found with item details (category, description, location, and optional private evidence like serial numbers for lost items).

2. **Automatic System Matching:** System searches for potential matches between lost and found items using key attributes. The system generates a suggested match list only, no automatic claim initiation occurs. The suggestion is presented to the administrator for review.
 - If potential matches found → Proceed to Step 5 (with admin review).
 - If no potential matches → Proceed to Step 3.
3. **Admin Checks Physical Inventory:** Admin reviews unmatched lost items and checks if the item was physically handed in but not logged. If available, admin logs it as found and proceeds to Step 5. If not, student is advised to wait.
4. **Security Logs Found Items:** Security personnel can log found items received physically, ensuring all items are entered into the system for matching.
5. **Notify Student and Initiate Claim:** Student is notified of a potential match and submits a claim, providing private proof of ownership (e.g., serial number, unique marks, photos of identical item) that is stored securely and only visible to administrators.
6. **Important Rule to Enforce (Verification):** Admin reviews the claim and private evidence to prevent fraudulent claims. The claim cannot proceed without admin approval. Evidence is stored in a separate secure table with restricted access.
7. **Approve and Schedule Handover:** Admin approves the claim. System updates status to resolved and creates a Chain of Custody Record with pickup time, location, and verification notes.
8. **Failure/Exception:** If admin rejects the claim (insufficient evidence or mismatch), status updates to rejected. Student is notified with reason, and the item remains available for other claimants.

Chain of Custody Record

Status	Description	Timestamp	Action By
lost	Item reported as lost by user	System timestamp	Student
found	Item logged as found (by security or matched)	System timestamp	Security/Admin
claimed	User submits claim with evidence	System timestamp	Student
verified	Admin reviews and verifies evidence	System timestamp	Administrator
resolved	Claim approved, handover completed	System timestamp	Administrator
rejected	Claim rejected due to insufficient evidence	System timestamp	Administrator
expired	Unclaimed after 30 days (disposed)	System timestamp	System

7. Design-suitability check

Check	Yes/No	Short evidence
Clear stakeholders and a genuine current problem	Yes	Section 4 details varied stakeholders (Students, Admins, and Security) and Section 1/2 outlines a genuine, inefficient manual process with specific observations and counts.
At least one state-sensitive workflow or entity	Yes	A lost/found item progresses through states: LOST → FOUND → CLAIMED → VERIFIED → RESOLVED/REJECTED. The claim verification step by an administrator (Section 7, Step 6) is critical to prevent fraud and ensure fair ownership transfer.
Important business/validation/security rule	Yes	Private evidence (serial numbers, unique marks) must remain confidential and be reviewed only by administrators. Automated matching provides suggestions only—no automatic claims.
Relevant quality trade-offs	Yes	Key trade-offs: usability (simple UI) vs. security (strict verification with private evidence), responsiveness vs. depth of data (search speed vs. detailed item descriptions), and transparency vs. privacy (public item listing vs. hidden ownership evidence).

Check	Yes/No	Short evidence
Relevant failure/exception condition	Yes	Claim rejection by admin is a clearly defined failure condition (Section 7, Step 8): If admin rejects the claim due to insufficient evidence or mismatch, status updates to REJECTED. Students are notified with reason, and the item remains available for other claimants.
Feasible semester-sized vertical slice	Yes	The core workflow (report, search suggestions, claim, approval, chain of custody) can be built as a functional web application within a semester.
Synthetic/anonymized data is sufficient	Yes	We can generate test data (e.g., sample lost/found items and user accounts) without needing real user data.
Not the University Service Hub or a renamed copy	Yes	The problem and solution are focused on general campus lost and found, not the lecturer case.

8. Data, ethics, safety and dependency risks

Risk/concern	How the team will control it
Data Privacy: System stores personal user data (name, email, student ID) and item details.	Implement RBAC and secure password hashing. Minimize stored personal data and anonymize test data. Include terms of use clarifying admin responsibility for physical verification. Private evidence (serial numbers, unique marks) stored in separate secure table with encrypted access limited to administrators only.
Fraudulent Claims: Malicious users may falsely claim expensive items.	Enforce administrative verification step (Section 7) requiring private evidence review before claim approval. Evidence remains private and is never displayed publicly or to other users.
System Dependencies: Relies on campus network, modern browser, and stable server.	Use widely supported technologies (Python/Flask, PostgreSQL). Design for robustness against network issues and graceful error handling. Document setup and deployment procedures.
Chain of Custody Integrity: Need to maintain accurate audit trail for all items.	Implement comprehensive logging of all status changes with timestamps and actor identification. Chain of Custody records are immutable (append-only) once created.

9. Lecturer decision

Decision	Select one
Approved	☐
Approved with minor conditions	☐
Revise and resubmit	☐

Field	Complete
Conditions/comments	
Lecturer	Dr Cleverence Kombe
Decision date / resubmission date if applicable	